

boAt Sales Performance & Business Intelligence Dashboard with Machine Learning

Developed Using: Power BI | Python | Pandas | Scikit-learn | DAX | Power Query

Executive Summary

The **boAt Sales Performance & Business Intelligence Dashboard** is an end-to-end analytics solution developed using **Power BI and Python**. The project transforms raw transactional sales data into meaningful business insights through interactive dashboards, statistical analysis, and machine learning. The solution enables stakeholders to monitor sales performance, product profitability, customer behavior, regional trends, and operational efficiency while also predicting potential loss-making orders using Machine Learning.

Project Objectives

The objectives of this project are:

- Analyze sales performance across different dimensions.
- Monitor profitability and revenue.
- Identify best-selling products.
- Analyze customer purchasing behavior.
- Evaluate regional sales performance.
- Perform Exploratory Data Analysis using Python.
- Build a Machine Learning model to predict loss-making orders.
- Integrate Machine Learning outputs into Power BI.

Tools & Technologies

Category	Tools
Business Intelligence	Power BI Desktop
Data Cleaning	Power Query
Programming	Python
Libraries	Pandas, NumPy, Matplotlib, Scikit-learn
Machine Learning	Random Forest Classifier
Data Modeling	Star Schema

Calculations	DAX
Version Control	GitHub

Data Cleaning & Preparation

Data cleaning was performed using **Python (Pandas)** and **Power Query**.

The following preprocessing steps were completed:

- Removed duplicate records.
- Checked missing values.
- Validated unique Product IDs.
- Standardized data types.
- Converted Order Date to Date format.
- Verified Discount Percentage.
- Calculated Product Margin.
- Validated Profit values.
- Checked Hero Products.
- Verified Festival records.
- Verified Cancellation and Return Reasons.
- Created Date hierarchy.

Data Modeling

A Star Schema was implemented.

Fact Table:

- Sales

Dimension Tables:

- Products
- Customers
- Geography
- Channels
- Delivery Partners
- Date

Relationships were established using primary and foreign keys to improve query performance.

Dashboard Development

The dashboard consists of **8 interactive report pages**.

Home

- Navigation Buttons
- Project Branding
- Dashboard Navigation

Executive Summary

KPIs

- Net Sales : `SUM('Sales_Data_Clean'[NetSales])`
- Profit : `SUM('Sales_Data_Clean'[Profit])`
- Orders : `DISTINCTCOUNT('Sales_Data_Clean'[SalesID])`
- Customers : `DISTINCTCOUNT('Sales_Data_Clean'[CustomerID])`
- Profit Margin : `DIVIDE([Total Profit],[Net Sales],0)`

Visuals

- Sales Trend
- Category Performance
- Profit by Category
- Sales by Channel

Sales Analysis

KPIs

- Gross Sales : `SUM('Sales_Data_Clean'[GrossMRPValue])`
- Quantity Sold : `SUM('Sales_Data_Clean'[Quantity])`
- Average Order Value : `DIVIDE([Net Sales],[Total Orders],0)`

Visuals

- Monthly Sales
- Profit Trend
- Sales by Channel
- Discount Analysis

Customer Analysis

KPIs

- Total Customers : `DISTINCTCOUNT('Sales_Data_Clean'[CustomerID])`
- Repeat Customers : `COUNTROWS(FILTER(VALUES('Sales_Data_Clean'[CustomerID]),CALCULATE(COUNTROWS('Sales_Data_Clean')) > 1))`
- Revenue per Customer : `DIVIDE([Net Sales],[Total Customers],0)`

Visuals

- Customer Segments
- Gender Analysis
- Tier Analysis
- Preferred Channels

Product Analysis

KPIs

- Products Sold : `DISTINCTCOUNT('Sales_Data_Clean'[ProductID])`
- Hero Product Sales : `CALCULATE([Net Sales],'Sales_Data_Clean'[HeroFlag] = 1)`
- Hero Product Profit : `CALCULATE([Total Profit],'Sales_Data_Clean'[HeroFlag] = 1)`

Visuals

- Product Category Sales
- Subcategory Analysis
- Launch Year Analysis
- Product Performance Table

Regional Performance

KPIs

- Total States : `DISTINCTCOUNT(Customer_Details_Clean[State])`
- Total Cities : `DISTINCTCOUNT(Customer_Details_Clean[City])`
- Orders : `DISTINCTCOUNT('Sales_Data_Clean'[SalesID])`
- Customers : `DISTINCTCOUNT('Sales_Data_Clean'[CustomerID])`

Visuals

- Sales by Region
- Sales by State
- Sales by City
- Regional Profit Distribution

Trend Analysis

KPIs

- Net Sales : `SUM('Sales_Data_Clean'[NetSales])`
- Gross Margin : `DIVIDE([Net Sales] - [Total COGS],[Net Sales],0)`
- Shipping Cost : `SUM('Sales_Data_Clean'[ShippingCost])`
- GST : `SUM('Sales_Data_Clean'[GSTAmount])`
- Return Rate : `DIVIDE([Returned Orders],[Total Orders],0)`

Visuals

- Profit Margin Waterfall
- Order Status
- Cancellation Analysis
- Python Histogram

Predictions & Machine Learning

KPIs

- Total Predictions : `COUNTROWS(Loss_Order_Predictions)`
- Correct Predictions :
`CALCULATE(COUNTROWS(Loss_Order_Predictions),Loss_Order_Predictions[Actual]=Loss_Order_Predictions[Predicted])`
- Incorrect Predictions : `[Total Predictions] - [Correct Predictions]`
- Prediction Accuracy : `DIVIDE([Correct Predictions],[Total Predictions])`

Visuals

- Predicted Order Distribution
- Feature Importance
- Actual vs Predicted Comparison
- Prediction Summary

Exploratory Data Analysis (Python)

EDA was performed using:

- Pandas
- NumPy
- Matplotlib

Analysis included:

- Missing Value Detection

- Duplicate Detection
- Statistical Summary
- Outlier Detection
- Correlation Analysis
- Distribution Analysis
- Profit Distribution
- Category-wise Analysis

Key observations were documented for each visualization.

Machine Learning Implementation

Business Problem

Predict whether an order will become a **loss-making order** before completion.

Algorithm Used

Random Forest Classifier

Workflow

- Feature Selection
- Train-Test Split (80:20)
- Model Training
- Prediction
- Model Evaluation
- Export Prediction Results
- Integrate Predictions into Power BI

Model Evaluation

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

The model achieved approximately **99% prediction accuracy** on the test dataset.

Key Business Insights

- True Wireless Earbuds generated the highest revenue.
- Hero products contributed the majority of overall sales.
- Tier-1 customers generated the highest revenue.
- Online Marketplace was the leading sales channel.
- Regional sales varied significantly across states.
- Approximately **29%** of historical orders were loss-making.
- Shipping cost and discount percentage significantly influenced profitability.
- Machine Learning accurately identified potential loss-making orders.

Business Recommendations

- Optimize discount strategies to reduce loss-making orders.
- Prioritize Hero Products during promotional campaigns.
- Focus marketing efforts on high-performing customer segments.
- Improve logistics to reduce shipping costs.
- Monitor regional performance regularly.
- Use predictive analytics to identify high-risk transactions before processing.
- Expand inventory for top-performing product categories.
- Develop targeted campaigns during festival periods.

Conclusion

This project demonstrates a complete Business Intelligence workflow by integrating data preparation, modeling, visualization, statistical analysis, and Machine Learning into a single solution. The dashboard enables business users to monitor key performance indicators, identify business opportunities, detect operational inefficiencies, and leverage predictive analytics for better decision-making. The integration of Power BI and Machine Learning provides a scalable framework for future business intelligence initiatives.

Screenshots





